

Australian statement of hazardous nature : Classified as hazardous according to criteria of Safe Work Australia

## Section 1 - Identification

**Product Name** Boron trifluoride, 12% in methanol

|                                |                                                                                                      |
|--------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Product Code</b>            | <b>445270000; 445271000; 445278000</b>                                                               |
| <b>Address</b>                 | ThermoFisher Scientific Australia Pty Ltd<br>5 Caribbean Drive, Scoresby<br>VICTORIA 3179, Australia |
| <b>Emergency Tel.</b>          | <b>CHEMTREC®</b><br><b>03 9757 4559 or +613 9757 4559</b>                                            |
| <b>Telephone / Fax Numbers</b> | Tel: 1300 735 292<br>Fax: 1800 067 639                                                               |
| <b>E-mail address</b>          | ANZinfo@thermofisher.com                                                                             |

**Recommended Use** Laboratory chemicals.

**Uses advised against** This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list. This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction. This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern.

## Section 2 - Hazard(s) Identification

### Classification under Safe Work Australia

Classified as hazardous according to criteria of Safe Work Australia

#### Physical hazards

Flammable liquids

Category 2

#### Health hazards

Acute Oral Toxicity  
Acute Dermal Toxicity  
Acute Inhalation Toxicity - Vapors  
Skin Corrosion/Irritation  
Serious Eye Damage/Eye Irritation  
Specific target organ toxicity - (single exposure)

Category 3  
Category 3  
Category 2  
Category 1 A  
Category 1  
Category 1

#### Environmental hazards

No hazards identified

### Label Elements



Flame



Skull and Crossbones



Health Hazard



Corrosion

**Signal Word****Danger****Hazard Statements**

H225 - Highly flammable liquid and vapor  
H314 - Causes severe skin burns and eye damage  
H330 - Fatal if inhaled  
H370 - Causes damage to organs  
H301 + H311 - Toxic if swallowed or in contact with skin  
AUH014 - Reacts violently with water

**Precautionary Statements**

P210 - Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
P233 - Keep container tightly closed  
P240 - Ground and bond container and receiving equipment  
P241 - Use explosion-proof electrical/ ventilating/ lighting equipment  
P242 - Use non-sparking tools  
P243 - Take action to prevent static discharges  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P284 - Wear respiratory protection  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P331 - Do NOT induce vomiting  
P363 - Wash contaminated clothing before reuse  
P370 + P378 - In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up  
P501 - Dispose of contents/ container to an approved waste disposal plant

**Other information**

Toxic to terrestrial vertebrates

## Section 3 - Composition and Information on Ingredients

| Component         | CAS No    | Weight % |
|-------------------|-----------|----------|
| Methyl alcohol    | 67-56-1   | 88       |
| Boron trifluoride | 7637-07-2 | 12       |

## Section 4 - First Aid Measures

|                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                          | Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Remove to fresh air. Immediate medical attention is required. If not breathing, give artificial respiration.                                                                                                                                     |
| <b>Ingestion</b>                           | Do NOT induce vomiting. Call a physician or poison control center immediately.                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Skin Contact</b>                        | Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                                        |
| <b>Eye Contact</b>                         | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                                                                                                                                                                  |
| <b>General Advice</b>                      | Immediate medical attention is required. Show this safety data sheet to the doctor in attendance.                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Self-Protection of the First Aider</b>  | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.                                                                                                                                                                                                                                                                                                                   |
| <b>First Aid Facilities</b>                | Eyewash, safety shower and washroom.                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b>Most important symptoms and effects</b> | Difficulty in breathing. Causes burns by all exposure routes. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting; Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: May cause blindness |
| <b>Notes to Physician</b>                  | Treat symptomatically. Symptoms may be delayed.                                                                                                                                                                                                                                                                                                                                                                                                                    |

## Section 5 - Fire Fighting Measures

### Suitable Extinguishing Media

CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam. Water mist may be used to cool closed containers.

### Extinguishing media which must not be used for safety reasons

Water.

### Hazardous Decomposition Products

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>), Oxides of boron, Gaseous hydrogen fluoride (HF), Thermal decomposition can lead to release of irritating gases and vapors.

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Reacts violently with water. Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back.

### Special protective equipment and precautions for fire fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## Section 6 - Accidental Release Measures

### Emergency procedures

Evacuate personnel to safe areas. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Ensure adequate ventilation. Remove all sources of ignition. Take precautionary measures against static discharges.

### Environmental Precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

**Methods for Containment and Clean Up****Clean-up methods - small spillage**

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Do not expose spill to water. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.

**Clean-up methods - large spillage**

Typically only supplied in small quantities as packaged goods.

If extremely toxic or used in larger quantities ensure a spillage action plan is in place. Evacuate area. Control the source and/or contain the spill if safe and able to do so. Use temporary diking, sand bags, dry sand, earth or proprietary booms/absorbent pads if available. Obtain advice on containment, neutralisation and clean-up from local emergency responders.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

**Section 7 - Handling and Storage****Precautions for Safe Handling**

Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Do not ingest. If swallowed then seek immediate medical assistance. Wear personal protective equipment/face protection. Do not allow contact with water. Keep away from open flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Conditions for Safe Storage, Including any Incompatibilities**

Keep away from heat, sparks and flame. Keep away from water or moist air. Refrigerator/flammables. Keep under nitrogen. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area.

AS/NZS 2243.10:2004, Safety in laboratories - Storage of chemicals

AS 1940-2004 - The storage and handling of flammable and combustible liquids

**Section 8 - Exposure Controls and Personal Protection****Exposure limits**

**AUS** - Exposure Standards for Atmospheric Contaminants in the Occupational Environment - Guidance Note on the Interpretation of Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:3008(1995)]

Adopted National Exposure Standards for Atmospheric Contaminants in the Occupational Environment [NOHSC:1003(1995)]

updated in August, 2005. Safe Work Australia **ACGIH** - Threshold Limit Values - Ceiling (TLV-C) guidelines by the American Conference of Governmental Industrial Hygienists (ACGIH) for controlling worker exposure to airborne chemical concentrations in the workplace. **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **DE** - MAK and BAT values of Hazardous Chemical Compounds in the Work Area. Published by German Research Foundation on July 1, 2011

| Component         | Australia                                                                                  | New Zealand WEL                                                                                    | ACGIH TLV                                                   | The United Kingdom                                                                                              | Germany                                                                                                                                                                                                                                |
|-------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl alcohol    | STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup> | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>Skin | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin                       | WEL - TWA: 200 ppm<br>TWA: 266 mg/m <sup>3</sup> TWA<br>WEL - STEL: 250 ppm<br>STEL: 333 mg/m <sup>3</sup> STEL | 100 ppm TWA MAK;<br>130 mg/m <sup>3</sup> TWA<br>MAK Skin absorber                                                                                                                                                                     |
| Boron trifluoride | TWA: 2.5 mg/m <sup>3</sup>                                                                 | Ceiling: 1 ppm<br>Ceiling: 2.8 mg/m <sup>3</sup>                                                   | TWA: 0.1 ppm TWA: 2.5 mg/m <sup>3</sup><br>Ceiling: 0.7 ppm |                                                                                                                 | TWA: 0.35 ppm (8 Stunden). AGW - exposure factor 2<br>TWA: 1 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 2<br>TWA: 1 mg/m <sup>3</sup> (8 Stunden). AGW - exposure factor 4<br>TWA: 1 mg/m <sup>3</sup> (8 Stunden). MAK Haut |

**Biological limit values**

NZ - Substances assigned Biological Exposure Indices in the New Zealand Workplace Exposure Standards and Biological Exposure Indices (6th edition). New Zealand Department of Labor

| Component      | Australia | New Zealand                                   | European Union | United Kingdom | Germany                                                                                                                                     |
|----------------|-----------|-----------------------------------------------|----------------|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl alcohol |           | 15 mg/L (urine) end of shift (Methyl alcohol) |                |                | Methanol: 15 mg/L urine (end of shift )<br>Methanol: 15 mg/L urine (for long-term exposures: at the end of the shift after several shifts ) |

#### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location. Use explosion-proof electrical/ventilating/lighting equipment.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### Eye Protection

Goggles (Australian/New Zealand Standard AS/NZS 1337 - Eye protectors for Industrial applications)

##### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | AUS/NZ Standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-----------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | AS/NZS 2161     | (minimum requirement) |

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### Skin and body protection

Long sleeved clothing

##### Respiratory Protection

Use an AS/NZS 1716 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained in line with AS/NZS 1715 on the use and maintenance of respiratory protective devices

##### Recommended Filter type:

##### Recommended half mask:-

low boiling organic solvent Type AX Brown conforming to EN371 (or AUS/NZ equivalent)  
Valve filtering: EN405 or Half mask: EN140 plus filter, EN 141 (or AUS/NZ equivalent)  
When RPE is used a face piece Fit Test should be conducted

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

#### Environmental exposure controls

No information available.

## Section 9 - Physical and Chemical Properties

#### Information on basic physical and chemical properties

##### Appearance

Clear

##### Physical State

Liquid

##### Odor

Odorless

##### Odor Threshold

No data available

##### pH

No information available

|                                                |                                               |                                             |
|------------------------------------------------|-----------------------------------------------|---------------------------------------------|
| <b>Melting Point/Range</b>                     | -98 °C / -144.4 °F                            |                                             |
| <b>Softening Point</b>                         | No data available                             |                                             |
| <b>Boiling Point/Range</b>                     | 65 °C / 149 °F                                | @ 760 mmHg                                  |
| <b>Flash Point</b>                             | 4 °C / 39.2 °F                                | <b>Method</b> - No information available    |
| <b>Evaporation Rate</b>                        | No data available                             |                                             |
| <b>Flammability (solid,gas)</b>                | Not applicable                                | Liquid                                      |
| <b>Explosion Limits</b>                        | <b>Lower</b> 5.5 vol%<br><b>Upper</b> 44 vol% |                                             |
| <b>Vapor Pressure</b>                          | No data available                             |                                             |
| <b>Vapor Density</b>                           | No data available                             | (Air = 1.0)                                 |
| <b>Specific Gravity / Density</b>              | 0.870                                         |                                             |
| <b>Bulk Density</b>                            | Not applicable                                | Liquid                                      |
| <b>Water Solubility</b>                        | May decompose                                 |                                             |
| <b>Solubility in other solvents</b>            | No information available                      |                                             |
| <b>Partition Coefficient (n-octanol/water)</b> |                                               |                                             |
| <b>Component</b>                               | <b>log Pow</b>                                |                                             |
| Methyl alcohol                                 | -0.74                                         |                                             |
| <b>Autoignition Temperature</b>                | No data available                             |                                             |
| <b>Decomposition Temperature</b>               | No data available                             |                                             |
| <b>Viscosity</b>                               | No data available                             |                                             |
| <b>Explosive Properties</b>                    |                                               | Vapors may form explosive mixtures with air |
| <b>Oxidizing Properties</b>                    | No information available                      |                                             |
| <b>Other information</b>                       |                                               |                                             |
| <b>Molecular Formula</b>                       | B F3                                          |                                             |
| <b>Molecular Weight</b>                        | 67.81                                         |                                             |

## Section 10 - Stability and Reactivity

|                                         |                                                                                                                                                                                     |
|-----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Reactivity</b>                       | Yes                                                                                                                                                                                 |
| <b>Stability</b>                        | Moisture sensitive.                                                                                                                                                                 |
| <b>Conditions to Avoid</b>              | Incompatible products, Excess heat, Exposure to moist air or water, Exposure to moisture, Keep away from open flames, hot surfaces and sources of ignition.                         |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Peroxides, Metals, Alkali metals, Acids, Acid anhydrides, Acid chlorides, Water.                                                                           |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO). Carbon dioxide (CO <sub>2</sub> ). Oxides of boron. Gaseous hydrogen fluoride (HF). Thermal decomposition can lead to release of irritating gases and vapors. |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                                                                                                                            |

## Section 11 - Toxicological Information

### Information on Toxicological Effects

#### Product Information

|                            |            |
|----------------------------|------------|
| <b>(a) acute toxicity;</b> |            |
| <b>Oral</b>                | Category 3 |
| <b>Dermal</b>              | Category 3 |
| <b>Inhalation</b>          | Category 2 |

| Component         | LD50 Oral                      | LD50 Dermal                   | LC50 Inhalation               |
|-------------------|--------------------------------|-------------------------------|-------------------------------|
| Methyl alcohol    | LD50 = 1187 – 2769 mg/kg (Rat) | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h |
| Boron trifluoride |                                |                               | LC50 = 194 ppm ( Rat ) 4 h    |

(b) skin corrosion/irritation; Category 1 A

(c) serious eye damage/irritation; Category 1

(d) respiratory or skin sensitization;

Respiratory

No data available

Skin

No data available

| Component                        | Test method                                                       | Test species | Study result    |
|----------------------------------|-------------------------------------------------------------------|--------------|-----------------|
| Methyl alcohol<br>67-56-1 ( 88 ) | OECD Test Guideline 406<br>Guinea Pig Maximisation Test<br>(GPMT) | guinea pig   | non-sensitising |

(e) germ cell mutagenicity; No data available

(f) carcinogenicity;

No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

| Component                        | Test method             | Test species / Duration       | Study result           |
|----------------------------------|-------------------------|-------------------------------|------------------------|
| Methyl alcohol<br>67-56-1 ( 88 ) | OECD Test Guideline 416 | Rat / Inhalation 2 Generation | NOAEC = 1.3 mg/l (air) |

(h) STOT-single exposure;

Category 1

Results / Target organs

Optic nerve

Central nervous system (CNS)

(i) STOT-repeated exposure;

No data available

Target Organs

No information available.

(j) aspiration hazard;

No data available

Other Adverse Effects

The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed

Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. May cause blindness

## Section 12 - Ecological Information

### Ecotoxicity effects

| Component         | Freshwater Fish                               | Water Flea                                | Freshwater Algae | Microtox                                                                        |
|-------------------|-----------------------------------------------|-------------------------------------------|------------------|---------------------------------------------------------------------------------|
| Methyl alcohol    | Pimephales promelas:<br>LC50 > 10000 mg/L 96h | EC50 > 10000 mg/L 24h                     |                  | EC50 = 39000 mg/L 25 min<br>EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min |
| Boron trifluoride |                                               | EC50: = 21.3 mg/L, 48h<br>(Daphnia magna) |                  |                                                                                 |

### Persistence and Degradability

#### Persistence

Persistence is unlikely, based on information available.

| Component      | Degradability |
|----------------|---------------|
| Methyl alcohol | DT50 ~ 17.2d  |

|                                        |                                                                                                                                                                                              |                                      |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| 67-56-1 ( 88 )                         |                                                                                                                                                                                              | >94% after 20d                       |
| <b>Bioaccumulative Potential</b>       | Bioaccumulation is unlikely                                                                                                                                                                  |                                      |
| <b>Component</b>                       | <b>log Pow</b>                                                                                                                                                                               | <b>Bioconcentration factor (BCF)</b> |
| Methyl alcohol                         | -0.74                                                                                                                                                                                        | <10 dimensionless                    |
| <b>Mobility</b>                        | The product contains volatile organic compounds (VOC) which will evaporate easily from all surfaces. Will likely be mobile in the environment due to its volatility Disperses rapidly in air |                                      |
| <b>Endocrine Disruptor Information</b> | This product does not contain any known or suspected endocrine disruptors                                                                                                                    |                                      |
| <b>Persistent Organic Pollutant</b>    | This product does not contain any known or suspected substance                                                                                                                               |                                      |
| <b>Ozone Depletion Potential</b>       | This product does not contain any known or suspected substance                                                                                                                               |                                      |

## Section 13 - Disposal Considerations

|                                            |                                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Waste from Residues/Unused Products</b> | Do not allow into drains or watercourses or dispose of where ground or surface waters may be affected. Wastes, including emptied containers, are controlled wastes and should be disposed of in accordance with all federal, E.P.A., state and local regulations. Assure conformity with all applicable regulations.                                                                  |
| <b>Contaminated Packaging</b>              | Dispose of this container to hazardous or special waste collection point. Empty containers retain product residue, (liquid and/or vapor), and can be dangerous. Keep product and empty container away from heat and sources of ignition.                                                                                                                                              |
| <b>Other Information</b>                   | Chemical wastes should be disposed through a licensed commercial waste collection service. Waste codes should be assigned by the user based on the application for which the product was used. Do not flush to sewer. Can be landfilled or incinerated, when in compliance with local regulations. Do not empty into drains. Large amounts will affect pH and harm aquatic organisms. |

## Section 14 - Transport Information

### IMDG/IMO

|                                |                                            |
|--------------------------------|--------------------------------------------|
| <b>UN-No</b>                   | UN3286                                     |
| <b>Proper Shipping Name</b>    | Flammable liquid, toxic, corrosive, n.o.s. |
| <b>Technical Shipping Name</b> | Boron trifluoride, 12% (1.5M) in methanol  |
| <b>Hazard Class</b>            | 3                                          |
| <b>Subsidiary Hazard Class</b> | 6.1 8                                      |
| <b>Packing Group</b>           | II                                         |

### ADG

|                                |                                            |
|--------------------------------|--------------------------------------------|
| <b>UN-No</b>                   | UN3286                                     |
| <b>Proper Shipping Name</b>    | Flammable liquid, toxic, corrosive, n.o.s. |
| <b>Technical Shipping Name</b> | Boron trifluoride, 12% (1.5M) in methanol  |
| <b>Hazard Class</b>            | 3                                          |
| <b>Subsidiary Hazard Class</b> | 6.1, 8                                     |
| <b>Packing Group</b>           | II                                         |

| Component                             | Hazchem Code |
|---------------------------------------|--------------|
| Methyl alcohol<br>67-56-1 ( 88 )      | 2WE          |
| Boron trifluoride<br>7637-07-2 ( 12 ) | 2RE          |

### IATA

|                                |                                            |
|--------------------------------|--------------------------------------------|
| <b>UN-No</b>                   | UN3286                                     |
| <b>Proper Shipping Name</b>    | Flammable liquid, toxic, corrosive, n.o.s. |
| <b>Technical Shipping Name</b> | Boron trifluoride, 12% (1.5M) in methanol  |

Hazard Class 3  
 Subsidiary Hazard Class 6.1 8  
 Packing Group II

Environmental hazards No hazards identified

Special Precautions No special precautions required

Additional information None known

## Section 15 - Regulatory Information

### Safety, health and environmental regulations/legislation specific for the substance or mixture

#### National Regulations Australia

See section 8 for national exposure control parameters.

#### **Standard for the Uniform Scheduling of Medicines and Poisons**

Classified as a scheduled poison according to the Standard for Uniform Scheduling of Medicines and Poisons.

| Component                     | Standard for the Uniform Scheduling of Medicines and Poisons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Methyl alcohol - 67-56-1      | Schedule 5 listed - except its derivatives; in preparations except a) when included in Schedule 10, or b) in preparations containing $\leq 2\%$ of Methanol, or c) when Methanol is present only as a denaturant of Ethanol<br>Schedule 6 listed - except its derivatives; except a) when included in Schedule 5, or b) when included in Schedule 10, or c) in preparations containing $\leq 2\%$ of Methanol<br>Schedule 10 listed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Boron trifluoride - 7637-07-2 | Schedule 2 listed<br>Schedule 3 listed<br>Schedule 4 listed - in preparations for human use except when included in or expressly excluded from Schedule 2 or 3<br>Schedule 4 listed - for human therapeutic use except: in preparations for internal use containing $\leq 6$ mg Boron per recommended daily dose, in preparations for dermal use containing $\leq 0.35\%$ of Boron, which are not for paediatric or antifungal use, or when present as an excipient<br>Schedule 5 listed - in preparations<br>Schedule 5 listed - in preparations except: in preparations for human use, or in preparations containing $\leq 15$ mg/kg of Fluoride ion; as Fluoride ion<br>Schedule 6 listed - in preparations except when included in Schedule 5<br>Schedule 6 listed - except: when included in Schedule 5, in preparations for human use, or in preparations containing $\leq 15$ mg/kg of Fluoride ion<br>Schedule 7 listed |

#### **Australian Industrial Chemicals Introduction Scheme (AICIS)**

| Component                     | Australian Industrial Chemicals Introduction Scheme (AICIS) | Additional information |
|-------------------------------|-------------------------------------------------------------|------------------------|
| Methyl alcohol - 67-56-1      | Present                                                     | -                      |
| Boron trifluoride - 7637-07-2 | Present                                                     | -                      |

#### **Australian - Illicit Drug Precursors/Reagents Substance List**

This product does not contain any substance(s) on the Illicit Drug Precursors/Reagents list.

#### **Chemicals of Security Concern**

This product does not contain any substance(s) listed on the voluntary National Code of Practice for Chemicals of Security Concern

National pollutant inventory Subject to reporting requirements

| Component | National pollutant inventory |
|-----------|------------------------------|
|-----------|------------------------------|

|                          |                                   |
|--------------------------|-----------------------------------|
| Methyl alcohol - 67-56-1 | 10 tonne/yr. Threshold category 1 |
|--------------------------|-----------------------------------|

**Prohibition or notification/licensing requirements**

Shown below are details of specific prohibition/notifications or licensing requirements when they apply.

This product does not contain any substance(s) subject to Prohibition, Authorization or Restriction.

**International Inventories**

| Component         | AICS | NZIoC | EINECS    | ELINCS | TSCA | DSL | NDSL | PICCS | ENCS | ISHL | IECSC | KECL     |
|-------------------|------|-------|-----------|--------|------|-----|------|-------|------|------|-------|----------|
| Methyl alcohol    | X    | X     | 200-659-6 | -      | X    | X   | -    | X     | X    | X    | X     | KE-23193 |
| Boron trifluoride | X    | X     | 231-569-5 | -      | X    | X   | -    | X     | X    | X    | X     | KE-03541 |

**Legend:** X - Listed. '-' - Not Listed. **KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**International Regulations**

**Ozone Depletion Potential** This product does not contain any known or suspected substance

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Rotterdam Convention (PIC)** Not applicable

**Basel convention on the control of transboundary movements of hazardous wastes and their disposal**

Not applicable.

| Component         | CAS No    | OECD HPV | Restriction of Hazardous Substances (RoHS) | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-------------------|-----------|----------|--------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Methyl alcohol    | 67-56-1   | Listed   | Not applicable                             | 500 tonne                                                                                 | 5000 tonne                                                                               |
| Boron trifluoride | 7637-07-2 | Listed   | Not applicable                             | 5 tonne                                                                                   | 20 tonne                                                                                 |

**Authorisation/Restrictions according to EU REACH**

| Component         | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                      | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methyl alcohol    | -                                                                   | Use restricted. See item 69. (see link for restriction details)<br>Use restricted. See item 75. (see link for restriction details) | -                                                                                                     |
| Boron trifluoride | -                                                                   | Use restricted. See item 75. (see link for restriction details)                                                                    | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Section 16 - Other Information****Legend**

|                                                                                                      |                                                                                                                              |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|
| <b>AICS</b> - Australian Inventory of Chemical Substances                                            | <b>NZIoC</b> - New Zealand Inventory of Chemicals                                                                            |
| <b>TSCA</b> - United States Toxic Substances Control Act Section 8(b) Inventory                      | <b>EINECS/ELINCS</b> - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances |
| <b>DSL/NDL</b> - Canadian Domestic Substances List/Non-Domestic Substances List                      | <b>ENCS</b> - Japanese Existing and New Chemical Substances                                                                  |
| <b>IECSC</b> - Chinese Inventory of Existing Chemical Substances                                     | <b>KECL</b> - Korean Existing and Evaluated Chemical Substances                                                              |
| <b>PICCS</b> - Philippines Inventory of Chemicals and Chemical Substances                            | <b>CAS</b> - Chemical Abstracts Service                                                                                      |
| <b>TWA</b> - Time Weighted Average                                                                   | <b>ACGIH</b> - American Conference of Governmental Industrial Hygienists Predicted No Effect Concentration (PNEC)            |
| <b>IARC</b> - International Agency for Research on Cancer                                            | <b>IMO/IMDG</b> - International Maritime Organization/International Maritime Dangerous Goods Code                            |
| <b>ICAO/IATA</b> - International Civil Aviation Organization/International Air Transport Association | <b>ADG</b> Australian Code for the Transport of Dangerous Goods by Road and Rail                                             |
| <b>MARPOL</b> - International Convention for the Prevention of Pollution from Ships                  | <b>OECD</b> - Organisation for Economic Co-operation and Development                                                         |
| <b>NZS 5433:2012</b> - Transport of Dangerous Goods on Land                                          | <b>LC50</b> - Lethal Concentration 50%                                                                                       |
| <b>LD50</b> - Lethal Dose 50%                                                                        | <b>ATE</b> - Acute Toxicity Estimate                                                                                         |
| <b>EC50</b> - Effective Concentration 50%                                                            | <b>RPE</b> - Respiratory Protective Equipment                                                                                |
| <b>WEL</b> - Workplace Exposure Limit                                                                | <b>NOEC</b> - No Observed Effect Concentration                                                                               |
| <b>DNEL</b> - Derived No Effect Level                                                                | <b>BCF</b> - Bioconcentration factor                                                                                         |
| <b>POW</b> - Partition coefficient Octanol:Water                                                     | <b>PBT</b> - Persistent, Bioaccumulative, Toxic                                                                              |
| <b>vPvB</b> - very Persistent, very Bioaccumulative                                                  |                                                                                                                              |
| <b>VOC</b> - (Volatile Organic Compound)                                                             |                                                                                                                              |

**Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Training Advice**

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

**Revision Date** 17-Nov-2022  
**Revision Summary** Not applicable.

**This Safety Data Sheet (SDS) is prepared in accordance to and complies with the requirements of Safe Work Australia - Work Health and Safety Regulations (WHS Regulations).**

**Disclaimer**

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## End of Safety Data Sheet